

DAY 2: PYTHON PRACTICE (MEAN & OUTLIERS)

Bilingual Practical Guide (English & Hindi) | Mission 2029

1. Basic Mean Calculation (Manual Method)

In this method, we calculate the mean using basic logic: sum of all items divided by the total number of items.

Is tareeke mein hum basic logic ka istemal karke mean nikalte hain: sabhi items ka jod aur unki kul ginti se bhag.

```
ring_weights = [5.4, 4.8, 6.2, 5.0, 5.5, 5.9, 4.5]
total_weight = sum(ring_weights)
number_of_rings = len(ring_weights)
mean_weight = total_weight / number_of_rings

print(f"Total Weight: {total_weight}g")
print(f"Mean Weight: {mean_weight:.2f}g")
```

Total Weight: 37.3g Mean Weight: 5.33g

2. Using 'Statistics' Library

Python provides a built-in library called 'statistics' to find the mean directly.

Python humein 'statistics' naam ki ek library deta hai jisse hum seedhe mean nikal sakte hain.

```
import statistics
staff_count = [12, 15, 14, 12, 15, 13]
avg_staff = statistics.mean(staff_count)
print(f"Average Daily Staff: {avg_staff}")
```

Average Daily Staff: 13.5

3. Identifying Outliers (Potential Issues)

If the mean value looks abnormally high compared to the rest of the data, it indicates an outlier.

Agar mean baki data ke mukable bahut bada dikhayi de, toh iska matlab data mein outlier hai.

```
DATA = [22, 24, 23, 22, 25, 100, 23]
mean_with_outlier = statistics.mean(DATA)
print(f"Mean with Outlier: {mean_with_outlier}")
```

Mean with Outlier: 34.142857142857146

4. Finding & Removing the Largest Value (The Max Logic)

To fix the data, we first find the maximum value (the outlier) and remove it from the list.

Data ko sahi karne ke liye, hum pehle sabse badi value (max) dhoondhte hain aur use list se hata dete hain.

```
# Step 1: Find the largest value
sabse_bada = max(DATA)
print(f"Largest Value: {sabse_bada}")

# Step 2: Remove the largest value and calculate new mean
DATA.remove(max(DATA))
new_mean = statistics.mean(DATA)
print(f"Corrected Data: {DATA}")
print(f"Mean after removing Outlier: {new_mean}")
```

Largest Value: 100 Corrected Data: [22, 24, 23, 22, 25, 23] Mean after removing Outlier: 23.166666666666668

Professional Analyst Tip: Always compare the Mean with and without outliers to understand their impact on your business reports.

Professional Analyst Tip: Hamesha outlier ke sath aur uske bina Mean ko compare karein taaki aap business reports par unke asar ko samajh sakein.